

LESSON 4-3 • SESSION 1 • PRACTICE • FORM B**Estimate with Benchmarks**

Learning target – I can estimate a percent by comparing an amount to benchmark percents. • 6.RP.A.3c

REMEMBER

- A **benchmark** is an easy percent: 10%, 25%, 50%, 75%, 100%.
- First name **the whole** (100%). Then find benchmark amounts: half for 50%, $\div 4$ for 25%, $\div 10$ for 10%.
- Choose the benchmark amount **closest** to the part. For the percent left, subtract from 100%.

WORKED EXAMPLE

A game download is 80 MB. 41 MB are done. About what percent is done?

Step 1 Whole: 80 MB is 100%.

Step 2 50% of 80 is 40 MB.

Step 3 41 MB is just over 40 MB.

Answer about 50%

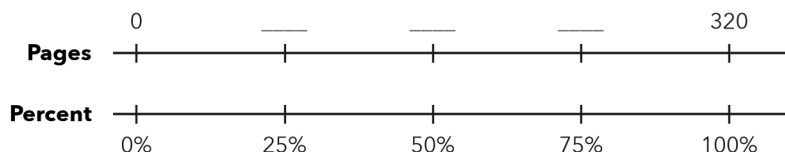
WORDS **estimate** (*estimar*) • **benchmark percent** (*porcentaje de referencia*) • **the whole** (*el todo*)

PRACTICE

show your work • label units

- About what percent? Name the closest benchmark: 10%, 25%, 50%, 75%, or 100%.
 - 29 out of 60 $\approx$ ____ %
 - 21 out of 80 $\approx$ ____ %
 - 46 out of 60 $\approx$ ____ %
 - 11 out of 100 $\approx$ ____ %
 - 39 out of 40 $\approx$ ____ %
 - 59 out of 120 $\approx$ ____ %
- Reading Goal.** Malik's goal is 320 pages this month. He has read 162 pages. Fill in the double number line.

Hint Each jump on the top line is $320 \div 4$ pages.



- Fill in the missing page numbers.
 - About what percent of his goal has he read? _____ %
- Class Fundraiser.** A class wants to raise \$600. So far it has raised \$452.

Hint Find 75% of \$600 and compare.

 - 10% of the goal _____
 - About what percent? _____ %
 - Gym Water Jug.** A water jug holds 48 cups. It has 23 cups left.
 - Half of the jug, in cups _____
 - About what percent is left? _____ %
 - Trail Progress.** A bike trail is 40 km long. A rider has gone 30 km.

Hint Find one fourth of 40 km first.

 - About what percent is done? _____ %
 - What percent is left? _____ %

Show your work.

- 6. Estimate Check.** Nia says 11 out of 40 is about 50%. Is she right? Use 25% of 40 to find a better estimate.

Hint Find 50% of 40 and 25% of 40.

Show your work.

Better estimate _____

- 7. Parking Lot.** A parking garage has 120 spaces. 89 are taken. Estimate the percent taken and name the benchmark amount you used.

Show your work.

About _____

- 8. Band Tickets.** A band printed 360 tickets for its show. It has sold 37 of them.

Hint Divide 360 by 10 to find 10%.

- a.** About what percent are sold? _____ %
- b.** About what percent are not sold? _____ %

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Marcus's Work.** A drive holds 400 GB. Marcus has used 300 GB, and 40 GB of that is photos. He says photos are about 75% of the drive. Explain his mistake. Then estimate the percent of the drive that is photos.

TEST PRACTICE

choose the best answer

- 10.** 50% of an amount is the same as –
- A** one tenth **B** one half **C** one fourth **D** twice it
- 11.** 9 out of 40 is about –
- A** 10% **B** 25% **C** 50% **D** 75%

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

24 C	25% E	\$60 E	37% Q	75% R	10% N	90% T	\$6 Z	50% P	70% X
#2b		#3a	#3b	#4a	#5b	#8a	#8b		

CHECK YOURSELF

☐ I can do this on my own
☐ I need a little help
☐ I have a question about #____